

EPISODE 0.1

To solve $(x + 3)(x + 5)(x + 7)(x + 9) = 9$.
(without expanding into a quartic polynomial equation at start)

Reference YouTube video and contributions in the comments:

https://www.youtube.com/watch?v=8CJ8n_hLoFI&list=WL&index=21

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Thinking Process 1

Mean of 3, 5, 7, 9 = 6

Let's try to apply this sense of symmetry [so as to see if we can apply $(x + a)(x - a) = x^2 - a^2$]:

$$[(x + 6) - 3][(x + 6) - 1][(x + 6) + 1][(x + 6) + 3] = 9$$

Let $y = x + 6$.

Then, the equation becomes

$$(y - 3)(y - 1)(y + 1)(y + 3) = 9$$

$$(y^2 - 1)(y^2 - 9) = 9$$

$$y^4 - 10y^2 = 0$$

$$y^2(y^2 - 10) = 0$$

$$y^2 = 0 \text{ or } y^2 = 10$$

$$y = 0 \text{ (repeated) or } y = \sqrt{10} \text{ or } y = -\sqrt{10}$$

When $y = 0$,

$$x + 6 = 0$$

$$x = -6$$

When $y = \sqrt{10}$,

$$x + 6 = \sqrt{10}$$

$$x = -6 + \sqrt{10}$$

When $y = -\sqrt{10}$,

$$x + 6 = -\sqrt{10}$$

$$x = -6 - \sqrt{10}$$

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Thinking Process 2a

Rearranging the factors with the sense that $3 + 9 = 12 = 5 + 7$.

Let's try to rearrange (so that coefficient of x^2 term = 1 and coefficient of x term = 12):

$$[(x + 3)(x + 9)][(x + 5)(x + 7)] = 9$$

$$(x^2 + 12x + 27)(x^2 + 12x + 35) = 9$$

Let $y = x^2 + 12x + 27$.

Then, the equation becomes

$$y(y + 8) = 9$$

$$y^2 + 8y - 9 = 0$$

$$(y - 1)(y + 9) = 0$$

$$y = 1 \text{ or } y = -9$$

When $y = 1$,

$$x^2 + 12x + 27 = 1$$

$$x^2 + 12x + 26 = 0$$

$$x = \frac{-12 \pm \sqrt{12^2 - 4(1)(26)}}{2(1)}$$

$$x = -6 \pm \sqrt{10}$$

When $y = -9$,

$$x^2 + 12x + 27 = -9$$

$$x^2 + 12x + 36 = 0$$

$$(x + 6)^2 = 0$$

$$x = -6 \text{ (repeated)}$$

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Thinking Process 2b

Rearranging the factors with the sense that $3 + 9 = 12 = 5 + 7$.

Let's try to rearrange (so that coefficient of x^2 term = 1 and coefficient of x term = 12):

$$[(x + 3)(x + 9)][(x + 5)(x + 7)] = 9$$

$$(x^2 + 12x + 27)(x^2 + 12x + 35) = 9$$

Let $t = x^2 + 12x$.

Then, the equation becomes

$$(t + 27)(t + 35) = 9$$

$$t^2 + 62t + 945 = 9$$

$$t^2 + 62t + 936 = 0$$

$$(t + 26)(t + 36) = 0$$

$$t = -26 \text{ or } t = -36$$

When $t = -26$,

$$x^2 + 12x = -26$$

$$x^2 + 12x + 26 = 0$$

$$x = \frac{-12 \pm \sqrt{12^2 - 4(1)(26)}}{2(1)}$$

$$x = -6 \pm \sqrt{10}$$

When $t = -36$,

$$x^2 + 12x = -36$$

$$x^2 + 12x + 36 = 0$$

$$(x + 6)^2 = 0$$

$$x = -6 \text{ (repeated)}$$

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Thinking Process 3

By considering the two numbers 5 and 7 as anchor points and consider $3 = 5 - 2$ and $9 = 7 + 2$ (which follows certain degree of symmetry):

$$[(x + 5) - 2](x + 5)(x + 7)[(x + 7) + 2] = 9$$

Let $a = x + 5$ and let $b = x + 7$.

Then, the equation becomes (see if we can get an equation in ab)

$$(a - 2)ab(b + 2) = 9$$

$$ab(a - 2)(b + 2) = 9$$

$$ab(ab + 2a - 2b - 4) = 9$$

$$ab[ab + 2(a - b) - 4] = 9$$

$$ab[ab + 2(-2) - 4] = 9$$

$$(ab)^2 - 8ab - 9 = 0$$

$$(ab - 9)(ab + 1) = 0$$

$$ab = 9 \text{ or } ab = -1$$

When $ab = 9$,

$$(x + 5)(x + 7) = 9$$

$$x^2 + 12x + 35 = 9$$

$$x^2 + 12x + 26 = 0$$

$$x = \frac{-12 \pm \sqrt{12^2 - 4(1)(26)}}{2(1)}$$

$$x = -6 \pm \sqrt{10}$$

When $ab = -1$,

$$(x + 5)(x + 7) = -1$$

$$x^2 + 12x + 35 = -1$$

$$x^2 + 12x + 36 = 0$$

$$(x + 6)^2 = 0$$

$$x = -6 \text{ (repeated)}$$